

## Atomic Structure and Radioactivity

1. The word "atom" is derived from the Greek word "Atomos," which means:
  - A) Tiny particle
  - B) Invisible
  - **C) Indivisible**
  - D) Hard sphere
2. The subatomic particle with a negative charge is the:
  - A) Proton
  - **B) Electron**
  - C) Neutron
  - D) Positron
3. The mass of an atom is primarily concentrated in the:
  - A) Shells
  - **B) Nucleus**
  - C) Orbitals
  - D) Electrons
4. Who discovered the nucleus through the Gold Foil experiment?
  - A) J.J. Thomson
  - B) John Dalton
  - **C) Ernest Rutherford**
  - D) Neils Bohr
5. The maximum number of electrons that can be accommodated in the 'L' shell is:
  - A) 2
  - **B) 8**
  - C) 18
  - D) 32
6. Neutrons were discovered by:
  - A) Rutherford
  - **B) James Chadwick**
  - C) Dalton
  - D) Goldstein
7. The atomic number (Z) represents the number of:

- **A) Protons**
  - B) Neutrons
  - C) Protons + Neutrons
  - D) Electrons in the outermost shell
8. **Isotopes are atoms of the same element with different numbers of:**
- A) Protons
  - B) Electrons
  - **C) Neutrons**
  - D) Positrons
9. **The formula to find the maximum electrons in a shell is:**
- A)  $n^2$
  - **B)  $2n^2$**
  - C)  $2n$
  - D)  $n+2$
10. **Which particle is nearly 1836 times lighter than a proton?**
- A) Neutron
  - **B) Electron**
  - C) Alpha particle
  - D) Hydrogen nucleus

## **Section 2: Radioactivity and Nuclear Science**

11. **Natural radioactivity was discovered by:**
- A) Marie Curie
  - **B) Henri Becquerel**
  - C) Ernest Rutherford
  - D) Enrico Fermi
12. **Which of the following radiations has the highest ionizing power?**
- **A) Alpha ( $\alpha$ ) rays**
  - B) Beta ( $\beta$ ) rays
  - C) Gamma ( $\gamma$ ) rays
  - D) X-rays
13. **Which radiation consists of high-energy electromagnetic waves?**
- A) Alpha
  - B) Beta

- **C) Gamma**
  - D) Cathode rays
14. **The process in which a heavy nucleus splits into two lighter nuclei is called:**
- **A) Nuclear Fission**
  - B) Nuclear Fusion
  - C) Chain Reaction
  - D) Radioactive Decay
15. **The isotope used for the treatment of cancer is:**
- A) Iodine-131
  - **B) Cobalt-60**
  - C) Carbon-14
  - D) Uranium-235
16. **Carbon-14 dating is used to determine the age of:**
- A) Rocks
  - **B) Fossils**
  - C) Planets
  - D) Ocean water
17. **Alpha particles are identical to the nucleus of:**
- A) Hydrogen
  - **B) Helium**
  - C) Lithium
  - D) Carbon
18. **The SI unit of radioactivity is the:**
- A) Curie
  - **B) Becquerel**
  - C) Roentgen
  - D) Rad
19. **Nuclear Fusion occurs naturally in:**
- A) Nuclear reactors
  - **B) The Sun and Stars**
  - C) Volcanic eruptions
  - D) Chemical laboratories
20. **Which material is commonly used as a moderator in nuclear reactors?**
- A) Iron
  - B) Lead

- **C) Graphite or Heavy Water**
  - D) Boron
21. **The mass of a neutron is approximately equal to the mass of a:**
- **A) Proton**
  - B) Electron
  - C) Alpha particle
  - D) Photon
22. **According to Bohr's model, electrons revolve in fixed paths called:**
- A) Clouds
  - B) Valencies
  - **C) Orbits/Shells**
  - D) Nuclei
23. **The 'M' shell can hold a maximum of how many electrons?**
- A) 2
  - B) 8
  - **C) 18**
  - D) 32
24. **A neutral atom of Carbon ( ${}_{6}^{12}\text{C}$ ) contains:**
- A) 6 protons, 12 neutrons
  - **B) 6 protons, 6 neutrons, 6 electrons**
  - C) 12 protons, 6 electrons
  - D) 6 protons, 12 electrons
25. **The distribution of electrons in different shells is known as:**
- A) Atomic number
  - B) Valency
  - **C) Electronic Configuration**
  - D) Ionization
26. **Which of the following is an example of an isotope of Hydrogen?**
- A) Deuterium
  - B) Protium
  - C) Tritium
  - **D) All of the above**
27. **The chemical properties of an element depend upon the number of:**

- A) Neutrons
  - **B) Valence electrons**
  - C) Protons in the nucleus
  - D) Total mass
28. **An atom becomes a positive ion (cation) by:**
- **A) Losing electrons**
  - B) Gaining electrons
  - C) Gaining protons
  - D) Losing neutrons
29. **The atomic mass (A) of an element is the sum of:**
- A) Protons and electrons
  - **B) Protons and neutrons**
  - C) Electrons and neutrons
  - D) Protons only
30. **Which shell is closest to the nucleus?**
- **A) K-shell**
  - B) L-shell
  - C) M-shell
  - D) N-shell
31. **The time required for half of the radioactive nuclei to decay is called:**
- A) Mean life
  - **B) Half-life**
  - C) Decay constant
  - D) Full life
32. **Beta (beta) particles are actually:**
- A) Protons
  - **B) Fast-moving electrons**
  - C) Helium nuclei
  - D) Neutrons
33. **Which radiation is the most penetrating?**
- A) Alpha
  - B) Beta
  - **C) Gamma**
  - D) UV rays
34. **Radioactive isotopes used in medicine are called:**
- **A) Radiopharmaceuticals**

- B) Biological tracers
  - C) Chemical catalysts
  - D) Nuclear fuels
- 35. **The process of estimating the age of non-living things like rocks is:**
  - A) Carbon dating
  - **B) Uranium dating**
  - C) Nitrogen dating
  - D) Fossil dating
- 36. **In a nuclear reactor, the "Coolant" is used to:**
  - A) Stop the reaction
  - **B) Absorb and transfer heat**
  - C) Increase the speed of neutrons
  - D) Shield radiation
- 37. **Which isotope is used to check the functioning of the thyroid gland?**
  - **A) Iodine-131**
  - B) Phosphorus-32
  - C) Carbon-12
  - D) Iron-59
- 38. **The fuel used in most nuclear power plants is:**
  - A) Coal
  - B) Hydrogen
  - **C) Uranium-235**
  - D) Helium
- 39. **Alpha decay reduces the atomic number of an element by:**
  - **A) 2**
  - B) 4
  - C) 1
  - D) 0
- 40. **Gamma rays have a charge of:**
  - A) +2
  - B) -1
  - **C) 0 (Neutral)**
  - D) +1
- 41. **The charge on an alpha particle is:**
  - **A) Positive**

- B) Negative
  - C) Neutral
  - D) Variable
42. **The number of neutrons in Tritium ( ${}^3_1\text{H}$ ) is:**
- A) 1
  - **B) 2**
  - C) 3
  - D) 0
43. **Which ray is used to sterilize medical instruments?**
- A) Alpha
  - B) Beta
  - **C) Gamma**
  - D) Cathode
44. **A device used to detect radiation is the:**
- **A) Geiger-Muller Counter**
  - B) Barometer
  - C) Voltmeter
  - D) Galvanometer
45. **When a Beta particle is emitted, the atomic number:**
- **A) Increases by 1**
  - B) Decreases by 1
  - C) Stays the same
  - D) Decreases by 2
46. **What is the mass number of an atom with 8 protons and 9 neutrons?**
- A) 8
  - B) 9
  - **C) 17**
  - D) 1
47. **Isotopes have the same:**
- **A) Chemical properties**
  - B) Physical properties
  - C) Atomic mass
  - D) Number of neutrons
48. **In the notation  ${}^A_Z\text{X}$ , 'A' stands for:**
- A) Atomic number
  - **B) Mass number**

- C) Atomic volume
- D) Avogadro's number

49. **Heavy water is:**

- A) H<sub>2</sub>O
- **B) D<sub>2</sub>O**
- C) T<sub>2</sub>O
- D) CO<sub>2</sub>

50. **Nuclear waste is usually buried in:**

- A) Forests
- B) Rivers
- **C) Deep underground / Salt mines**
- D) Atmosphere

51. **Which scientist compared the atom to "plum pudding"?**

- A) Neils Bohr
- **B) J.J. Thomson**
- C) John Dalton
- D) James Chadwick

52. **The region around the nucleus where the probability of finding an electron is high:**

- A) Shell
- **B) Orbital**
- C) Path
- D) Core

53. **In the 'N' shell, the value of 'n' is:**

- A) 1
- B) 2
- C) 3
- **D) 4**

54. **If an atom has 11 protons and 12 neutrons, its atomic number is:**

- **A) 11**
- B) 12
- C) 23
- D) 1

55. **The simplest atom that contains no neutrons in its most common isotope is:**

- **A) Hydrogen**

- B) Helium
  - C) Lithium
  - D) Beryllium
56. **Cathode rays are composed of:**
- A) Protons
  - **B) Electrons**
  - C) Neutrons
  - D) Positrons
57. **Canal rays (Positive rays) were discovered by:**
- A) J.J. Thomson
  - **B) Goldstein**
  - C) Dalton
  - D) Bohr
58. **The valency of an atom with 7 electrons in its outermost shell is usually:**
- **A) 1**
  - B) 7
  - C) 8
  - D) 0
59. **Which isotope is used to study the process of photosynthesis?**
- A) Carbon-12
  - **B) Carbon-14**
  - C) Iodine-131
  - D) Cobalt-60
60. **An alpha particle carries a charge of:**
- A) +1
  - **B) +2**
  - C) -1
  - D) 0

## **Section 7: Nuclear Energy & Radiation Hazards**

61. **The conversion of a stable nucleus into an unstable one by bombardment is:**
- A) Natural radioactivity
  - **B) Artificial radioactivity**
  - C) Fission

- D) Fusion
- 62. **The most dangerous radiation if accidentally swallowed or inhaled is:**
  - **A) Alpha** (due to high ionization)
  - B) Beta
  - C) Gamma
  - D) X-rays
- 63. **Which of the following is used as a safety shield in nuclear plants?**
  - A) Aluminum
  - B) Iron
  - **C) Lead and Concrete**
  - D) Wood
- 64. **The energy released in the Sun is due to:**
  - A) Fission of Uranium
  - **B) Fusion of Hydrogen into Helium**
  - C) Burning of Coal
  - D) Chemical combustion
- 65. **A chain reaction is controlled in a nuclear reactor using:**
  - A) Graphite rods
  - **B) Boron or Cadmium rods**
  - C) Iron rods
  - D) Water pipes
- 66. **Radioactive isotopes used to trace the path of an element in a system are:**
  - **A) Tracers**
  - B) Moderators
  - C) Coolants
  - D) Shields
- 67. **The mass of an alpha particle is approximately:**
  - A) 1 amu
  - B) 2 amu
  - **C) 4 amu**
  - D) 0 amu
- 68. **Which rays are not deflected by electric or magnetic fields?**
  - A) Alpha rays
  - B) Beta rays

- **C) Gamma rays**
  - D) Cathode rays
69. **Half-life of Carbon-14 is approximately:**
- A) 100 years
  - B) 1500 years
  - **C) 5730 years**
  - D) 25000 years
70. **The unit of absorbed radiation dose is:**
- A) Curie
  - **B) Gray (or Rad)**
  - C) Becquerel
  - D) Newton

### Section 8: Isotopes & Detailed Properties

71. **Deuterium is an isotope of Hydrogen containing:**
- **A) 1 proton, 1 neutron**
  - B) 1 proton, 2 neutrons
  - C) 2 protons, 1 neutron
  - D) 1 proton, 0 neutrons
72. **The sum of the number of protons and neutrons is the:**
- A) Atomic number
  - **B) Mass number**
  - C) Relative atomic mass
  - D) Valency
73. **Atoms of different elements having the same mass number are:**
- A) Isotopes
  - **B) Isobars**
  - C) Isotones
  - D) Isomers
74. **The maximum number of electrons in the 'K' shell is:**
- **A) 2**
  - B) 8
  - C) 18
  - D) 32
75. **Which radiation has the least ionizing power?**
- A) Alpha

- B) Beta
  - **C) Gamma**
  - D) X-ray
76. **Radioactivity is a property of the:**
- A) Electrons
  - B) Shells
  - **C) Nucleus**
  - D) Chemical bonds
77. **An electron has a mass of approximately:**
- **A)  $9.11 \times 10^{-31}$  kg**
  - B)  $1.67 \times 10^{-27}$  kg
  - C)  $1.67 \times 10^{-24}$  kg
  - D)  $9.11 \times 10^{-27}$  kg
78. **When an atom loses an electron, it becomes:**
- **A) Positively charged**
  - B) Negatively charged
  - C) Neutral
  - D) Radioactive
79. **Isotopes of Uranium used in nuclear reactors is:**
- A) U-238
  - **B) U-235**
  - C) U-234
  - D) U-236
80. **Nuclear fusion requires very high:**
- A) Pressure only
  - **B) Temperature**
  - C) Humidity
  - D) Darkness

## **Section 9: Applied Science & Review**

81. **The isotope used to detect leaks in underground pipes is:**
- **A) Sodium-24**
  - B) Iodine-131
  - C) Carbon-14
  - D) Cobalt-60
82. **A beta particle is represented as:**
- A)  $_{+2}^4\text{He}$

- **B)  $_{-1}^{0}\text{e}$**
  - C)  $_{+1}^{1}\text{H}$
  - D)  $_{0}^{1}\text{n}$
83. **Which shell has the highest energy?**
- A) K
  - B) L
  - C) M
  - **D) N** (among these four)
84. **The discovery of the electron proved that the atom is:**
- A) Indivisible
  - **B) Divisible**
  - C) Solid
  - D) Square-shaped
85. **The central part of the atom is called:**
- A) Proton
  - B) Electron
  - **C) Nucleus**
  - D) Neutron
86. **In an atom, the number of protons is always equal to the number of:**
- A) Neutrons
  - **B) Electrons** (in a neutral atom)
  - C) Positrons
  - D) Alpha particles
87. **The isotopes of Chlorine are:**
- **A) Cl-35 and Cl-37**
  - B) Cl-34 and Cl-36
  - C) Cl-35 and Cl-36
  - D) Cl-37 and Cl-38
88. **The radioactive gas used in the treatment of cancer is:**
- A) Helium
  - B) Neon
  - **C) Radon**
  - D) Argon
89. **The splitting of a nucleus was first observed by:**
- **A) Otto Hahn and Fritz Strassman**
  - B) Newton

- C) Einstein
  - D) Dalton
90. **Alpha rays consist of a stream of:**
- A) Electrons
  - B) Protons
  - **C) Helium nuclei**
  - D) Neutrons

## **Section 10: Final Review**

91. **What is the charge of a neutron?**
- **A) 0**
  - B) +1
  - C) -1
  - D) +2
92. **The flow of charge is called:**
- **A) Current**
  - B) Voltage
  - C) Resistance
  - D) Power
93. **Radioactivity is a:**
- A) Chemical change
  - **B) Nuclear change**
  - C) Physical change
  - D) Biological change
94. **The most common isotope of Hydrogen is:**
- **A) Protium**
  - B) Deuterium
  - C) Tritium
  - D) Helium
95. **Valency of noble gases (like Neon) is:**
- **A) 0**
  - B) 1
  - C) 2
  - D) 8
96. **Which rays have the highest velocity (equal to light)?**
- A) Alpha
  - B) Beta

- **C) Gamma**
  - D) Cathode
97. **Isotopes have different:**
- A) Atomic numbers
  - **B) Physical properties**
  - C) Chemical properties
  - D) Electron counts
98. **The moderator in a reactor slows down:**
- A) Electrons
  - B) Protons
  - **C) Neutrons**
  - D) Atoms
99. **An example of a natural radioactive element is:**
- A) Iron
  - B) Gold
  - **C) Uranium**
  - D) Aluminum
100. **The mass number is written as a:**
- A) Subscript \* **B) Superscript** \* C) Prefix \* D) Suffix